

# STM32 LoRa and SENSOR Input with ADC Circuit

With Analog Front and SMA Sensor Connector

## Power Delivery

USB Type C

SMPS (Digital)  
+5v to 3v3

LDO (Analog)  
+5v to 3v3

Bias Generator  
(Op-Amp) 1v65



## Microcontroller + LoRa

HSE

TIMER

LORA

I2C

UART

SWD

ADC

## LoRa matching/filter network

Output Filter

Output Matching

TX

Input Filter

Input Matching

RX

## ADC Circuit

BUFFER

ANTI ALIASING  
FILTER

## SENSORS

MPU -6050  
(IMU)

BME280 (Temp,  
Humidity + Pressure)

## Connectors and Breakout Pins

SWD Progammer

I2C +UART

BNC ADC SENSOR

BNC LoRa Antenna

Analog Front End

SMA Sensor  
Connector

